

Zero-Shot and Fine-Tuned Evaluation of Generative LLMs for Arabic Word Sense Disambiguation

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Abstract

Arabic presents unique challenges for sense-level language understanding due to its rich morphology and semantic ambiguity. This paper benchmarks large generative language models (LLMs) for Arabic Word Sense Disambiguation (WSD) under both zero-shot and fine-tuning conditions. We evaluate one proprietary model (GPT-4o) and three open-source models (LLaMA 3.1-8B, Qwen 2.5-7B, and Gemma 2-9B) on two publicly available datasets. In zero-shot settings, GPT-4o achieved the highest overall performance, with comparable results across both datasets, reaching 79% accuracy and an average macro-F1 score of 66.08%. Fine-tuning, however, notably elevated all open models beyond GPT-4o’s zero-shot results. Qwen achieved the top scores on one dataset, with an accuracy of 90.77% and a macro-F1 score of 83.98%, while LLaMA scored highest on the other, reaching an accuracy of 88.51% and a macro-F1 score of 69.41%. These findings demonstrate that parameter-efficient supervised adaptation can close much of the performance gap and establish strong, reproducible baselines for Arabic WSD using open-source, relatively medium-sized models. Full code is publicly available.¹

1 Introduction

Word Sense Disambiguation (WSD) is a core problem in Natural Language Processing (NLP) that involves determining which sense of a word is intended within a particular context. This task is especially challenging due to semantic polysemy, where individual words can convey multiple meanings depending on their context of use. Arabic, in particular, significantly amplifies this complexity due to its rich morphological structure and substantial polysemy (Al-Hajj and Jarrar, 2021; Kaddoura and Nassar, 2024b).

فَقَدْ نَفْسَ عَزِيزٍ عَلَيْهِ أَصْدَرَ الْقَاضِي نَفْسَ الْحَكْمِ

(1) (2)

A representative example is the word (نَفْس, *nafs*), which has different meanings depending on the context. In sentence (1), (*faqada nafs ‘azīzin ‘alayhi*), meaning (*he lost a dear soul*), the word refers to (soul). In sentence (2), (*aṣḍara al-qāḍī nafs al-ḥukm*), meaning (*the judge issued the same ruling*), it means (same).

Furthermore, omitting diacritics in written Arabic exacerbates ambiguity, complicating the task of accurate disambiguation (Alqahtani et al., 2019).

Before the advent of modern Artificial Intelligence (AI) methods, traditional approaches dominated WSD tasks. These older methods primarily involved rule-based strategies utilizing lexical databases and glossaries, alongside statistical and dictionary-based approaches (Abey Siriwardana and Sumanathilaka, 2024; Eid et al., 2010). Although foundational, these traditional methodologies exhibited limitations in scalability and contextual adaptability.

Recent advancements in NLP have introduced powerful Large Language Models (LLMs) that significantly enhance the ability to address semantic tasks through the use of contextualized representations. Encoder-based models, such as BERT (Devlin et al., 2019), have demonstrated high effectiveness in various language understanding tasks through supervised fine-tuning on labeled data. In Arabic, adaptations such as AraBERT (Antoun et al., 2020) and CAMELBERT (Inoue et al., 2021) have enabled the capture of Arabic linguistic features more effectively.

On the generative side, autoregressive decoder-based models such as the GPT series (Radford et al., 2018), and newer multilingual and Arabic-capable models like LLaMA (Touvron et al., 2023), Owen (Bai et al., 2023), Jais and Jais-chat (Sen-

¹<https://github.com/Yosrranour1996/Arabic-WSD-LLM>

gupta et al., 2023), and ALLaM (Bari et al., 2024) have opened the door for zero-shot, few-shot, and fine-tuning-based learning approaches. Unlike encoder-based architectures, these generative models operate by predicting the next token in a sequence, making them well-suited for prompt-based inference and instruction-following settings. This architectural distinction underpins differences in how each family of models performs disambiguation, offering complementary strengths for WSD evaluation.

Despite these advancements, the potential of generative LLMs for Arabic WSD remains not well explored. In this paper, we provide the following contributions:

- **We analyze** available Arabic WSD datasets and identify those most suitable for evaluation.
- **We evaluate** generative LLMs under zero-shot and fine-tuned settings, assessing their effectiveness in Arabic sense disambiguation.

2 Related Work

The recent rise of Pre-trained Language Models (PLMs) has significantly advanced NLP, leading to extensive efforts to benchmark their effectiveness across diverse linguistic contexts and a wide range of Arabic NLP tasks.

For instance, ORCA (Elmadany et al., 2023) introduced a benchmark covering 60 Arabic Natural Language Understanding (NLU) datasets across seven tasks, including WSD. Using the dataset by El-Razzaz et al. (2021), they reported a top F1-score of 76.68% with AraBERTv2, highlighting its effectiveness in MSA-based disambiguation.

Moreover, GPTAraEval (Khondaker et al., 2023) extended the evaluation to dialectal Arabic, revealing significant performance gaps between Modern Standard Arabic (MSA) and dialectal varieties when assessed using ChatGPT (GPT-3.5) and GPT-4. For WSD, they also utilized the dataset by El-Razzaz et al. (2021), in which ChatGPT achieved a best F1-score of 53.49% in a three-shot setting, reflecting the limitations of general-purpose LLMs in fine-grained disambiguation.

More recently, AraReasoner (Hasanaath et al., 2025) conducted a broad evaluation of reasoning-oriented LLMs, including DeepSeek models, across fifteen Arabic NLP tasks using various prompting and fine-tuning strategies. On the

same dataset, their fine-tuned DeepSeek-R1-Q 14B model achieved up to 86.27% F1 score, demonstrating the effectiveness of task-specific adaptation.

In parallel, the ArabicNLP 2024 shared task (Khalilia et al., 2024) evaluated WSD systems on the SALMA dataset (Jarrar et al., 2023). The baseline model, a Target Sense Verification (TSV) system with a context window of 11 words, achieved the highest accuracy of 84.2%. Among the participants, Upaya obtained a top result of 77.82% using LLaMA-3-70B-Instruct with structural prompting. The shared task also evaluated Location Mention Disambiguation (LMD) using the IDRISI-DA dataset (Suwaileh et al., 2023a,b), which was created in two phases—first extracting location mentions, then disambiguating them. In this task, systems retrieved and reranked candidate toponyms from OpenStreetMap, with the best model achieving MRR@1 of 0.95.

Furthermore, EnhancedBERT (Kaddoura and Nassar, 2024b) introduces an ensemble BERT approach for Arabic WSD offering complementary benchmark.

Several other studies have also explored the performance of LLMs on Arabic NLP. However, most of these focus on specific applications or broader task suites that exclude WSD. In some cases, researchers develop their own datasets and conduct evaluations within that scope. Still, these efforts often lack generalization to fine-grained sense disambiguation, leaving essential gaps in systematic evaluation.

3 Arabic WSD Datasets: A Review

This section reviews key datasets for Arabic WSD, referred to here as Dataset A to Dataset F, with a focus on their construction methods and annotation schemes, summarized in Table 1.

Dataset A. Proposed by El-Razzaz et al. (2021), this dataset addresses the shortage of Arabic gloss-based resources by providing a public benchmark consisting of 15,549 senses for 5,347 unique Arabic words, extracted from the Modern Standard Arabic Dictionary. It frames Arabic WSD as a binary classification task, distinguishing between correct and incorrect glosses for a given word-in-context.

Dataset B. Proposed by Jarrar et al. (2023), SALMA is a novel Arabic sense-annotated cor-

Aspect	Dataset A	Dataset B	Dataset C	Dataset D	Dataset E	Dataset F
Corpus size	15.5K tokens	34K tokens	3.7K sentence	27.5K sentence	28K pairs	167K pairs
Coverage	Single lemmas	Single lemmas	100 polysemous words	Single lemmas	Single lemmas	Single lemmas
Annotation	Gloss binary	Relatedness scores	Sense labeling	Gloss-based	Gloss binary	Gloss true/false
Construction method	Semi-Automatic	Manual	Manual, GPT-3.5	Fully Manual	Semi-Automatic	Semi-Automatic
Data type/Domain	Dictionary examples	News and Media	Multi-domain	Multi-domain	Dictionary examples	Arabic Ont., Lex.
Source	MSA dictionary	Modern, Ghani	Web, GPT-3.5	DHDA dictionary	CAD dictionary	Arabic Ont., Lex.

Table 1: Summary of major Arabic WSD datasets (A–F). Abbrev.: Ont. = Ontology; Lex. = lexicography

pus containing around 34K tokens (approximately 29K annotated words), annotated simultaneously using two lexicons (Modern and Ghani). Unlike traditional binary methods, SALMA introduces a graded scoring system that assigns semantic relatedness scores to each sense (ranging from 1% to 100%). It also includes additional annotations for named entities.

Dataset C. Introduced by [Kaddoura and Nassar \(2024a\)](#), this dataset contains 3,670 context sentences representing 367 distinct senses across 100 carefully selected Arabic polysemous words. Sentences were manually collected from diverse online sources (e.g., news, medicine, finance) and supplemented with GPT-3.5-generated examples to cover less frequent senses.

Dataset D. Introduced by [Saidi et al. \(2023\)](#), WSDTN is a large-scale, manually annotated corpus of 27,530 Arabic sentences. It offers extensive semantic coverage. The annotation is based on the Doha Historical Dictionary of Arabic (DHDA).

Dataset E. Proposed at the KSAA-CAD shared task ([Alshammari et al., 2024](#)). This dataset provides approximately 28K Arabic gloss-context pairs sourced from the Contemporary Arabic Language Dictionary (CAD).

Dataset F. [Al-Hajj and Jarrar \(2021\)](#) introduced a significantly large dataset comprising approximately 167K context-gloss pairs extracted from the Arabic Ontology and the Birzeit lexicographic databases. The dataset is structured as a binary classification task (true/false).

4 Experimental Setup

4.1 Dataset Preparation

Dataset A ² and Dataset B ³ were selected for evaluation as they are publicly available and offer complementary properties in terms of size, annotation schemes, and sense granularity.

Formatting and Preprocessing. Both datasets were organized into a consistent format comprising: (i) context sentences including the target word and candidate senses, (ii) ground-truth sense labels, and (iii) a dictionary mapping sense IDs to glosses. For Dataset B, tokens with invalid POS tags or missing semantic annotations were filtered to ensure cleaner input for disambiguation, and the sense with the highest score was treated as the correct label. The formatting strategy followed an approach similar to that used in the ArabicNLP 2024 shared task ([Khalilia et al., 2024](#)). Dataset examples are available in Appendix A.

Train-Test Splits. As shown in Table 2, custom 64/16/20 partitions were constructed for both datasets. For Dataset A, which contains a single target token per sentence, a random sentence-level split was applied to create training, development, and test sets. For Dataset B, where sentences may contain multiple targets, stratification was performed at the token level to ensure that 80% of annotated tokens were allocated to training and development, and 20% to testing. We ensured that no sentence appeared in both splits, preventing data leakage.

²<https://github.com/MElrazzaz/Arabic-word-sense-disambiguation-bench-mark>

³<https://sina.birzeit.edu/salma/>

Dataset	Train	Dev	Test	Total
Dataset A	9952	2487	3,110	15,549
Dataset B	18427	4691	5,781	28,899

Table 2: Token-level split statistics for the selected WSD datasets. Since no official splits are provided, custom partitions were created.

4.2 Model Selection

Two categories of models were compared in the evaluation:

- **Open-source LLMs:** LLaMA 3.1-8B (Grattafiori et al., 2024), Qwen 2.5-7B (Qwen et al., 2025), and Gemma 2-9B (Team et al., 2024).
- **Proprietary LLM:** GPT-4o (OpenAI et al., 2024).

The open-source models were evaluated under both zero-shot prompting and supervised fine-tuning. GPT-4o is used exclusively in the zero-shot setting, as fine-tuning this model is currently not feasible. This setup enables us to assess the effectiveness of instruction-tuned models for Arabic WSD and to examine how well relatively compact LLMs (7B–9B) perform compared to larger proprietary systems.

Zero-Shot Prompt:	Fine-Tuning TE:
You are	{ <code>"instruction": "You are ..."</code> ,
Sentence: ...	<code>"input": "Sentence: '...' "</code>
Target Word: ...	<code>Target Word: '...' Possible "</code>
Possible Senses:	<code>Senses: [...]"</code> ,
- Sense ID:, Definition:	<code>"output": "correct sense "</code>
- Sense ID:, Definition:	<code>ID"]</code> }
Correct Sense ID is:	

Figure 1: Example formats for zero-shot prompting (left) and fine-tuning training examples (right).

4.3 Prompting and Fine-Tuning Strategies

Zero-Shot Prompting. All models were evaluated using a consistent prompt format that included the sentence, the target word, and a list of possible senses with their definitions. The models were instructed to select the correct sense ID (see Figure 1, left). Inference was performed using a mix of local deployments and API access, and we did not enforce deterministic decoding or temperature constraints. This stage aimed to assess how effectively large generative models can disambiguate senses in Arabic purely through instruction-following.

Supervised Fine-Tuning. For supervised adaptation, the open-source models were locally fine-tuned on the training splits of the benchmarks using parameter-efficient strategies, and specifically applied LoRA (Hu et al., 2021) in all experiments to reduce the memory footprint and training time. Training examples were formatted as instruction-style JSON objects containing the sentence, the target word, the candidate senses, and the correct label (see Figure 1, right). To handle long examples, truncation was applied non-uniformly: Dataset A samples were short, while for Dataset B sequences were retained up to 4,096 tokens, reducing training samples to 18,357. Training was performed on an NVIDIA L4 GPU with models loaded in 4-bit precision. Hyperparameters were tuned empirically to balance convergence speed and overfitting risk:

- **Dataset A:** epochs = 3, batch = 1 ($8 \times$ accumulation), lr = 2×10^{-4} , max_len = 1024, LoRA rank = 32, α = 32, dropout = 0.05, packing = True, eval_steps = 100
- **Dataset B:** epochs = 1, batch = 1 ($8 \times$ accumulation), lr = 2×10^{-4} , max_len = 4096, LoRA rank = 16, α = 16, dropout = 0.0, packing = False, eval_steps = 500
- **Common:** optimizer = AdamW_8bit, weight decay = 0.01, scheduler = linear, warmup steps = 50, gradient checkpointing = True, mixed precision = fp16/bf16 (auto), seed = 3407, load_in_4bit = True

4.4 Evaluation Metrics

Performance was evaluated using two complementary metrics: accuracy and macro-F1. Together, these measures provide a balanced perspective on how effectively the models addressed the task.

5 Results

Table 3 reports the results achieved by each model across both datasets.

5.1 Zero-Shot Results

In the zero-shot evaluation, GPT-4o achieved the highest performance, with similar results across both datasets. Among the open-source models, Gemma 2-9B performed best, particularly on Dataset B, where it surpassed the other models by a clear margin. Qwen 2.5-7B consistently outperformed LLaMA 3.1-8B, which had the lowest performance among the evaluated models.

Model	Dataset A				Dataset B			
	Zero-shot		Finetuning		Zero-shot		Finetuning	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Gemma 2-9B	65.34	50.64	89.39	81.72	72.46	56.45	87.23	67.80
LLaMA 3.1-8B	48.59	38.28	90.42	83.20	54.78	39.98	88.51	69.41
Qwen 2.5-7B	67.40	53.02	90.77	83.98	55.99	47.97	82.22	63.07
GPT-4o	79.16	67.92	—	—	79.55	64.23	—	—

Table 3: Accuracy and Macro-F1 scores of different models on Dataset A and Dataset B for Arabic WSD.

5.2 Fine-Tuning Results

Parameter-efficient fine-tuning led to substantial improvements across all open models. Qwen 2.5-7B achieved the best performance on Dataset A, while LLaMA-3.1-8B, despite its lower zero-shot results, improved markedly with supervised adaptation and reached the highest scores on Dataset B. Gemma 2-9B also demonstrated significant gains across both datasets.

5.3 Results Analysis

We summarize four recurring phenomena observed in both datasets; for concreteness, we illustrate the patterns with LLaMA (see Figure 2):

- **In-set “close” vs. “distant” errors.** Close errors arise when glosses are near-paraphrases, whereas distant errors reflect semantic divergence. For الإفرقيي, the gold “مَنْسُوبٌ إِلَى إِفْرِيقِيَا” was predicted as “اسم منسوب إلى إفريقييَا” a trivial close miss. In contrast, for هرب من “الحفلة بالنوم”, the gold “هرب من مسؤولياته: تنصل” was predicted as “هرب فلان منها، تملص منها”, a distant error. Zero-shot runs were dominated by distant errors, as the datasets contain relatively few close glosses, (e.g., Gemma has on Dataset B: 1,465 distant vs. 5 close). However, fine-tuning markedly reduced them (e.g., LLaMA on Dataset A: 938 distant errors reduced to 291).
- **Invalid outputs (refusals + hallucinations).** Zero-shot models sometimes refused or produced non-existent IDs; LLaMA had 638 refusals on Dataset A and 539 on Dataset B. After fine-tuning, invalid outputs disappeared. Qwen also dropped from 1,277 refusals on Dataset B to 261 after tuning.
- **Effect of sense inventory size and dataset style.** Accuracy falls as the candidate set grows. Dataset A (dictionary-style; mostly

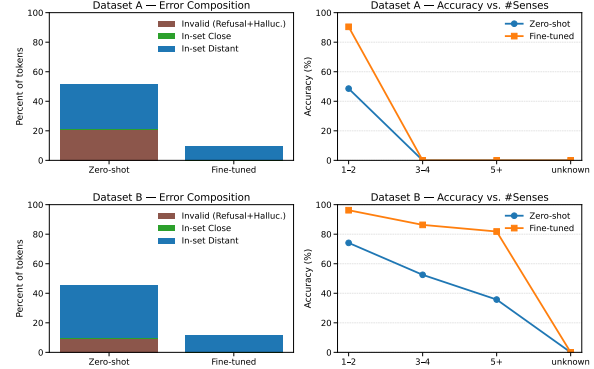


Figure 2: LLaMA-3.1-8B: zero-shot vs. fine-tuned on Dataset A and Dataset B. **Left:** 100% error composition (Invalid = refusal+hallucination; In-set Close; In-set Distant). **Right:** accuracy by number of candidate senses.

1–2 senses per token) is easier, whereas Dataset B (corpus-based; many items with 5+ senses) is harder.

- **Difficult tokens.** In zero-shot, Dataset B concentrated errors on proper nouns and abstract terms (e.g., الجوائز, المركزية), while Dataset A’s hardest cases were highly polysemous dictionary items (e.g., أمل, لبق). Fine-tuning removed zero-accuracy tokens in Dataset A, but some Dataset B tokens remained challenging (e.g., غرب).

6 Conclusion

This study benchmarked generative LLMs for Arabic WSD in zero-shot and fine-tuned settings across two public datasets. While GPT-4o led in zero-shot, parameter-efficient fine-tuning of open models consistently closed the gap and surpassed that baseline, yielding strong, reproducible results. Our analysis shows that factors such as sense-inventory size and error type drive performance differences and largely explain the gains from fine-tuning. Future work can expand to dialects.

Limitations

- **Dataset Scope.** This study focuses on two publicly available Modern Standard Arabic (MSA) datasets. The findings may not generalize to dialectal Arabic or other domains with different sense distributions and annotation practices.
- **Model Coverage.** We limited our evaluation to widely used multilingual LLMs. Arabic-centric models such as Jais and ALLaM, which may yield stronger performance, were not included due to stability and resource considerations.
- **Prompting Design.** To establish a clean zero-shot baseline, we used a minimal instruction-following prompt without examples or chain-of-thought reasoning. Richer prompting strategies (e.g., few-shot, reasoning heuristics, alternative gloss formats) could improve results but were left for future work.

Acknowledgments

We thank the Department of Electrical and Electronic Engineering, University of Khartoum, for their support.

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Appendix

A Dataset Examples

Figure 3 and Figure 4 show illustrative examples from Dataset A and Dataset B, respectively, including the test-set sentence, the dictionary sense mapping, and the gold label.

